From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

FIITJEE

JEE(Advanced)-2014

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

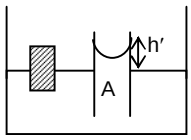
CRT –IV
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	B	C
2.	A	A	D
3.	B	D	A
4.	D	C	A
5.	C	C	A
6.	D	B	D
7.	D	A	A
8.	D	C	A
9.	B	A	D
10.	C	C	A
11.	B, C	C, D	A, B, D
12.	A, B, C	A, C	A, B, D
13.	A, B	A, B, C, D	A, B
14.	B, C	A, D	B, D
15.	A, D	A, D	A, D
1.	8	2	2
2.	0	3	2
3.	2	3	4
4.	4	3	1
5.	1	8	4

Physics

PART – I

1. There will be no change.
$\therefore \quad h' = h$



2. $\varepsilon = \dfrac{\sigma}{\varepsilon_0}$

$v_A - v_B = \dfrac{\sigma}{\varepsilon_0}(b) - \dfrac{\sigma}{\varepsilon_0}(a)$

$\therefore \; W = Q(v_A - v_B) = \dfrac{Q\sigma}{\varepsilon_0}(b-a)$

3. $\omega = 0 + 1 \times 10 = 10 \text{ rad/sec}^2$
$\therefore \quad v = r\omega = 1 \times 10 = 10 \text{ m/s}$

$\bar{B} = \dfrac{\mu_0}{4\pi}\dfrac{q(\bar{v}\times\bar{r})}{r^3} \quad \Rightarrow \; |\bar{B}| = \dfrac{\mu_0 qv}{4\pi r^2}$

$B = \dfrac{10^{-7}\times 0.1\times 10}{(1)^2} = 10^{-7}\text{ T}$

4. conservation of momentum

$M\sqrt{2gh} - m\sqrt{2gh} = MV_1 + mV_2$ (1)

$-1 = \dfrac{V_1 - V_2}{2\sqrt{2gh}}$ (2)

$V_2 = 3\sqrt{2gh}$

$h' = \dfrac{V_2^2}{2g} = 9h$

5. $\dfrac{1}{C} = \dfrac{1}{C_1} + \dfrac{1}{C_2}$

$= \dfrac{x}{\varepsilon_0 A} + \dfrac{a-b-x}{\varepsilon_0 A}$

$\Rightarrow \; C = \dfrac{\varepsilon_0 A}{a-b}$

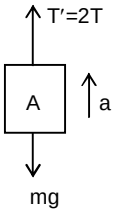
6. FBD of 'B' and 'C'



$\Rightarrow \; T - 2g = 2a$ . . . (i)
and $3g - T = 3a$ . . . (ii)

$T = \dfrac{12g}{5}$

for A



$\Rightarrow$ T = mg

$2 \times \frac{12g}{5} = mg \Rightarrow m = 4.8$ kg.

7. $W = \int_0^{\pi/2} f.Rd\theta = \frac{\mu mgR\pi}{2} = 1$ joule.

9. $a = \frac{\rho v(g+a) - m(g+a)}{m} = 20\ m/s^2$

$t = \sqrt{\frac{2h}{a}} = 1$ sec

11. Angular momentum = $\frac{nh}{2\pi}$

kinetic energy $\propto \frac{1}{n^2}$

De-broglie wavelength $\lambda = \frac{h}{mv}$ so as v increases λ decreases.

14. Relative velocity of B w.r.t. A along vertical direction should be zero

$20 \sin\theta = 10$

$\Rightarrow \sin\theta = 1/2$

$x = 20 \cos 30° \times \frac{1}{2} = 5\sqrt{3}$ m

15. Change in angular momentum = angular impulse

$L = 5\tau$

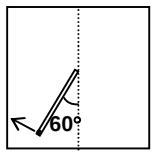
$I = \frac{ma^2}{3}$

$K = \frac{L^2}{2I} = \frac{75\tau^2}{2ma^2}$

Section -C

1. Conservation of energy $\frac{1}{2}mv^2 - mg\frac{\ell}{2} = mg\frac{\ell}{2} + m\sqrt{3}\left(2\ell\frac{\sqrt{3}}{2}\right)g$

$v = \sqrt{8g\ell}$



2. Potential across capacitor is zero, hence energy stored is zero.

3. $v_A - v_B = \int E.\,dx = 10\ (1)$

$= 10$ volt.

4. $Z = \sqrt{R^2 + X^2} = \sqrt{9 + X^2}$

but $\cos \phi = \frac{R}{Z} = \frac{3}{5}$

$X = 4\ \Omega$.

5. $\frac{E}{2} = E - ir$

$\Rightarrow \quad i = \frac{E}{2r}$ . . . (i)

$2E = i(3 + r)$ . . . (ii)

$\Rightarrow r = 1\ \Omega$

Chemistry

PART – II

SECTION – A

1. $K_c = 4 \times 10^{-19}$

$Ag^+ + 2CN^- \rightleftharpoons \left[Ag(CN)_2\right]^{-1}$

0.03 0.1 0

$\left[Ag(CN)_2\right]^{-1} \rightleftharpoons Ag^+ + 2CN^-$

$t = 0$	0.3	0	0.04
	$0.03 - x$	x	$0.04 + 2x$

$$K_c = \frac{x(0.04+2x)^2}{[0.03-x]} = 4\times10^{-19}$$

$0.04 + 2x \approx 0.04$

$0.03 - x \approx 0.03$

$$K_c = \frac{x.(0.04)^2}{0.03} = 4\times10^{-19}$$

$x = 7.5 \times 10^{-18}$

2. $PV = nRT$

As PV constant

$n_1T_1 = n_2T_2$

$n_1 . 300 = n_2 . 400$

$$n_2 = \frac{300}{400}n_1 = \frac{3}{4}n_1$$

4. Energy of $\overset{\ominus}{e}$ in H-atom = - E

Energy of $\overset{\ominus}{e}$ in Li^{+2} = - 9E

Energy supplied by photon = $IE + \frac{1}{2}mv^2$

$IE = |E|$

$$\therefore E_p = 9E + \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2(E_p - 9E)}{m}}$$

9. $FeSO_4 \xrightarrow{\Delta} Fe_2O_3 + SO_3 + SO_2$

12. $H^+ + e^- \longrightarrow \frac{1}{2}H_2 \qquad E^o = 0.0\ v$

$$E = E_{H^+/H_2} = 0 - \frac{0.059}{1}\log\frac{1}{[H^+]}$$

$= -0.059 \times 7 = -0.413$

So, $E_{H^+/H_2} < E_{Ni^{2+}/Ni}$

Ni will deposit at cathode

$2H_2O + 2e^- \longrightarrow H_2 + 2OH^-$

$E = -0.83 - \frac{0.059}{2}\log\left[OH^-\right]^2$

$= -0.83 + 0.059 \times 7$

= - 0.417

13. Both have $Cr^{+3}(d^3)$, and C.N. is 6. Hence d^2sp^3.

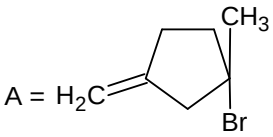
$Pd^{+2}(d^8)$ complexes with C.N. is 4 are square plannar and magnetic moment is zero.

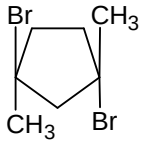
14. $FeCl_3 \xrightarrow{\Delta} FeCl_2 + Cl_2$

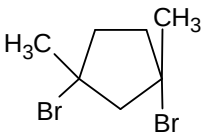
Dil. $CuCl_2$ is blue due to $[Cu(H_2O)_4]^{2+}$ PH_3 is obtained by heating white P with NaOH

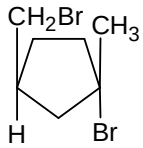
All are reducing because of weak P – H bond.

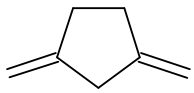
SECTION – C

1. A = 

C = 

B = 

E = 

D =  inactive

2. $KNO_3 \rightarrow KNO_2 + \frac{1}{2}O_2$

$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$

$NH_4NO_3 \rightarrow N_2O + 2H_2O$

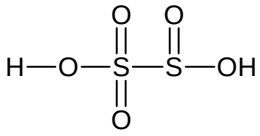
$2Ca(NO_3)_2 \rightarrow 2SrO + 4NO_2 + O_2$

$2Mg\left(NO_3\right)_2 \rightarrow 2MgO + 4NO_2 + O_2$

$NaNO_3 = NaNO_2 + \frac{1}{2}O_2$

$Ca\left(NO_3\right)_2$, $Sr\left(NO_3\right)_2$,

$Pb(NO_3)_2$ do not respond to the test.

3. 

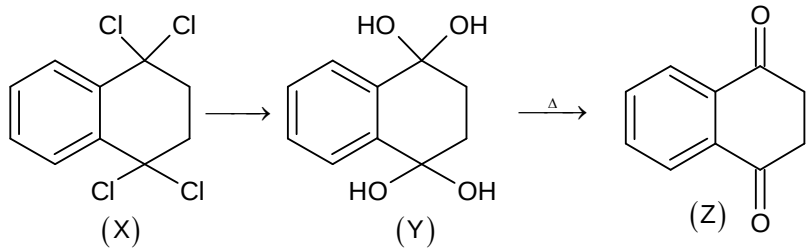
4. $\Delta T_b = K_b \times m \times i$

$2.08 = 0.52 \times 1 \times i$

$i = 4$

$K_3\left[Fe'''(CN)_6\right]$

5.



(X) (Y) (Z)

Mathematics

PART – III

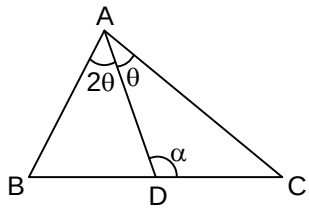
1. $I = \int_{-2}^{1} [x[1+(-1)]+1]\,dx + \int_{-1}^{1} [x[1+0]+1]dx$

2. Here, $g(x) = \begin{cases} f(x) & , \text{ if } 0 \le x < \frac{1}{2} \\ f\left(\frac{1}{2}\right) & , \text{ if } \frac{1}{2} \le x \le 1 \\ 3-x & , \text{ if } 1 < x \le 2 \end{cases}$

4. Let $\frac{A}{3} = \angle CAD = \theta$

By m – n theorem,

(1 + 1) cot α = 1 cot 2θ – 1cot θ



6. Here, $z_2 - z_3 = i(z_1 - z_3)$

$\Rightarrow \angle ACB = \frac{\pi}{2}$

$\Rightarrow AC = \frac{13}{\sqrt{2}}$; AC = BC

9. (0, 0) lies on the director circle

10. $\frac{x+1}{x^{2/3}-x^{1/3}+1} - \frac{x-1}{x-x^{1/2}} = x^{1/3} - x^{-1/2}$

12. $x + y + z = 3\left(\frac{a+b}{2}\right);\ \frac{1}{x}+\frac{1}{y}+\frac{1}{z} = \frac{3\left(\frac{1}{a}+\frac{1}{b}\right)}{2}$

14. Here, $f^{-1}(x) = a + \sqrt{x-a}$

SECTION – C

1. Equation of tangent is $y - \beta = f'(\alpha)\{x - \alpha\}$

Also, $\alpha = \frac{\beta - \alpha f'(\alpha)}{\sqrt{f'(\alpha)^2+1}}$

Thus, $\frac{dy}{dx} = \frac{1}{2}\left(\frac{y^2-x^2}{xy}\right)$

Thus, equation of curve is $x^2 + y^2 - 2x = 0$

2. Equation of normal is $y - a^n = -\frac{1}{na^{n-1}}(x-a)$

4. Here, $\Delta_1 = d(a_1 - a_2)(a_2 - a_3)(a_3 - a_1) = -2d^4$
Also, $\Delta_2 = -2d^4$

5. $\sin^2 x[\sin^2 x + \sin x + 2] = 0$
$\Rightarrow \sin x = 0$, where $x = 0, \pi, 2\pi, 3\pi$